1. When several processes access the same data concurrently and the outcome of the execution depends on the
order in which the access takes place, is called
A: dynamic condition
B: race condition
C: essential condition
D: critical condition

2. If a process is executing in its critical section, then no other processes can be executing in their critical section:
This condition is called
A: mutual exclusion
B: critical exclusion
C: synchronous exclusion
D: asynchronous exclusion

3. Mutual exclusion can be provided by the
A: mutex locks
B: binary semaphores
C: both A and B
D: none of the mentioned

4. When high priority task is indirectly preempted by medium priority task effectively inverting the relative priority of
the two tasks, the scenario is called
A: priority inversion
B: priority removal
C: Priority exchange
D: priority modification

5. Process synchronization can be done on
A: Hardware level
B: software level
C: both A and B
D: none of the mentioned

6. The interval from the time of submission of a process to the time of completion is termed as

A: waiting time
B: turnaround time
C: response time
D: throughput

7. In priority scheduling algorithm, when a process arrives at the ready queue, its priority is compared with the priority
of?
A: all process
B: currently running process
C: parent process
D: init process

8. Which one of the following cannot be scheduled by the kernel?
A: kernel level thread
B: user level thread
C: process
D: none of the mentioned

9. Which of the following condition is required for deadlock to be possible?
A: mutual exclusion
B: a process may hold allocated resources while awaiting assignment of other resources
C: no resource can be forcibly removed from a process holding it
D: all of the mentioned

10. A system is in the safe state if
A: the system can allocate resources to each process in some order and still avoid a deadlock
B: there exist a safe sequence
C: both A and B
D: none of the mentioned

11. What is the drawback of banker's algorithm?
A: in advance processes rarely know that how much resource they will need
B: the number of processes changes as time progresses
C: resource once available can disappear
D: all the mentioned

12. To avoid deadlock
A: there must be a fixed number of resources to allocate
B: resource allocation must be done only once.
C: all deadlocked processes must be aborted
D: inversion technique can be used

13. Multithreaded programs are:
A: lesser prone to deadlocks
B: more prone to deadlocks
C: not at all prone to deadlocks
D: None of these

14. For Mutual exclusion to prevail in the system:
A: at least one resource must be held in a non-sharable mode
B: the processor must be a uniprocessor rather than a multiprocessor
C: There must be at least one resource in sharable mode
D: All of these

15. Deadlock prevention is a set of methods:
A: to ensure that at least one of the necessary conditions cannot hold
B: to ensure that all the necessary conditions do not hold
C: to decide if the requested resources for a process must be given or not
D: to recover from a deadlock

16. To ensure that the hold and wait condition never occurs in the system, it must be ensured that:
A: whenever a resource is requested by a process, it is not holding any other resources
B: each process must request and be allocated all its resources before it begins its execution
C: a process can request resources only when it has none
D: All of these

17. A system is in a safe state only if there exists a:
A: safe allocation
B: safe resource
C: safe sequence
D: All of these

18. What is compaction?
A: a technique for overcoming internal fragmentation
B: a paging technique
C: a technique for overcoming external fragmentation
D: a technique for overcoming fatal error

19. Operating System maintains the page table for
A: each process
B: each thread
C: each instruction
D: each address

20. A process is thrashing if:
A: it spends a lot of time executing, rather than paging
B: it spends a lot of time paging, than executing
C: it has no memory allocated to it
D: None of these

21. Which file is a sequence of bytes organized into blocks understandable by the System 's linker?
A: object file
B: source file
C: executable file
D: text file

22. Mapping of file is managed by
A: file metadata
B: page table
C: virtual memory
D: file system

23. The time taken for the desired sector to rotate to the disk head is called:
A: Positioning time
B: Random Access time
C: seek time
D: Rotational Latency

24. _____________ writes occur in the order in which the disk subsystem receives them, and the writes are not
buffered.
A: Asynchronous
B: Regular
C: Synchronous
D. Irregular

25. A file being read or written sequentially should not have its pages replaced in LRU order, because
______________.
A: it is very costly
B: the most recently used page will be used last
C: it is not efficient
D: All of these

26. File system fragmentation occurs when
A: unused space or single file are not contiguous
B: used space is not contiguous
C: unused space is non-contiguous
D: multiple files are non-contiguous

27. When a page fault occurs before an executing instruction is complete:
A: the instruction must be restarted
B: the instruction must be ignored
C: the instruction must be completed ignoring the page fault
D: None of the mentioned

28. In virtual memory the programmer _________ of overlays.
A: has to take care
B: does not have to take care
C: None of these
D: Both A & B
29. When a page fault occurs, the state of the interrupted process is:
A: disrupted
B: invalid
C: saved
D: None of these

30. When a process begins execution with no page in memory:
A: process execution becomes impossible
B: a page fault occurs for every page brought into memory
C: process causes system crash
D: None of these

1. For which of the following offset can be positive or negative?
Answers
1. SEEK_SET
2. SEEK_END
3. SEEK_CUR--A
4. All of the above
5. None of the above

2. In which of the IPC mechanism, data is not copied from user space to kernel space and vice a versa?
Answers
1. Pipe
2. Message queue
3. Shared memory--A
4. socket

3. A bootloader is responsible for
i. loading an operating system kernel and its components
ii. loading supporting infrastructure into memory
iii. beginning the kernel's execution
Answers
1. i and ii
2. i and iii
3. ii and iii
4. All of the above--A

4. In which of the following state change in child process, performing wait allows the system to release
the resources associated with child process?
Answers
1. the child terminated--A

2. the child was stopped by a signal
3. the child was resumed by a signal
4. All of the above

5. Which of the following is not used to examine and change the signal action?
Answers
1. signal
2. sigaction
3. sigprocmask--A
4. All of the above

7. Select correct option for mutex.
Answers
1. A thread can lock mutex twice.
2. A thread locking mutex is owner of that mutex.--A
3. Owner can not unlock the mutex.
4. None of the above

8. What is internal fragmentation?
Answers
1. process is not utilizing the whole partition allocated to it.--A
2. process is utilizing the whole partition allocated to it.
3. amount of space required for process is not available.
4. amount of space required for process is available, but not contiguous.

9. Physical memory : ________________ :: Logical Memory : ________________
Answers
1. Pages, Frames
2. Frames, Pages--A
3. Pages, Fragments
4. fragments, Frames

10. If the size of logical address space is 2 to the power of m, and a page size is 2 to the power of n
addressing units, then the high order _____ bits of a logical address designate the page number,
and the ____ low order bits designate the page offset.
Answers

1. m, n
2. n, m
3. m – n, m
4. m – n, n--A

11. LRU page replacement algorithm suffers from Belady’s anomaly.
Answers
1. true
2. false--A

12. Which of the following is journaling file system
Answers
1. JFS
2. UFS
3. ext2
4. ext3--A

13. Thrashing
Answers
1. reduces page I/O
2. decreases the degree of multiprogramming
3. implies excessive page I/O--A
4. improves the system performance

14. While fork(), the child's set of pending signals is initially _____.
Answers
1. filled with same as parent
2. empty--A
3. filled except masked signals in parent
4. None of the above

15. The child does not inherit ________________.
Answers
1. semaphore adjustments from its parent
2. its parent's memory locks
3. timers from its parent
4. All of the above--A
5. None of the above

16. Which of the following architecture does not support embedded operating system
Answers
1. ARM
2. AVR32
3. MIPS
4. None of the above--A

17. ________________ provide the information about the existence of files, their location on secondary memory, their current status and other attributes.
1. Memory Table
2. I/O Table
3. File Tables--A
4. Process Tables

18.
```
#include <stdio.h>
#include <unistd.h>
int main()
{
        fork();
        fork();
        fork();
        printf(" A New Process Created.");
        return 0;
}
```
How many times Above message "A New Process Created" is printed?
Answers
1. 1
2. 3
3. 8--A
4. 16

19. sigprockmask() system call does ______________.
Answers
1. change the process signal mask.
2. retrive the existing mask
3. Both of the above--A
4. None of the above

20. Spinlocks are intended to provide ____________ only.
Answers
1. Mutual Exclusion
2. Bounded Waiting--A
3. Aging
4. Progress

21. Which of the following not belong to exec() family?
Answers
1. execv();
2. execvp();
3. execvpe();
4. execlv();--A

22. msgsnd() returns an integer. which of the following is true statement.
Answers
1. Return value > 1 indicates a correct send.
2. Return value = 0 indicates a correct send
3. Both of above--A
4. Return value = -1 indicates an error has occurred

23. _______________ is a technique of gradually increasing the priority of the processes that wait in the system for a long time.
Answers
1. Starvation
2. Waiting queue
3. Aging--A
4. Non of the above

24. Multiple source files are compiled together to form a single kernel binary image. Such a kernel called as ______________.
Answers
1. Micro-kernel
2. Monolithic kernel--A
3. Modular kernel
4. Hybrid kernel

25. Named pipe or FIFO can be created by ___________ command.
Answers
1. pipe
2. mkfifo--A
3. mkpipe
4. makefifo

26. Bankers algorithm is an example of _________________
Answers
1. deadlock prevention
2. deadlock avoidance--A
3. deadlock detection
4. deadlock recovery

27. Preemption is ______________
Answers
1. forced deallocation of the CPU from a program which is executing on the CPU--A
2. release of CPU by a program after the completing its task
3. forced allotment of CPU by a program to itself
4. a program is terminating itself due to detection of error

28. Which one of the following bootloader is not used by linux?
Answers
1. GRUB
2. LILO
3. NTLDR--A
4. None of the mentioned

29. Each thread has its own user stack and no kernel stack.
Answers
1. True
2. False--A

30. Thread synchronization is required because ___________
Answers
1. all threads of a process share the same address space
2. all threads of a process share the same global variables
3. all threads of a process can share the same files

4. all of the mentioned--A

31. Mutex Functionality :
Answers
1. based up on locking mechanism--A
2. based up on signalling mechanism
3. both A and B
4. None of the above

32. On success, pthread_join() returns :
Answers
1. 0--A
2. 1
3. Error No
4. None of the above

33. fork() returns non zero value in child process and zero in parent process.
Answers
1. False--A
2. True

34. Select odd option from below
Answers
1. execl("./cmdline", "cmdline", "one", "two", "three", "four", NULL);
2. char *args[] = { "cmdline", "one", "two", "three", NULL }; execv("./cmdline", args);
3. execlp("ps", "ps", "-e", "-o", "pid,ppid,cmd");--A
4. None of the above

35. Which is Fastest IPC mechanism
Answers
1. FIFO
2. Pipe
3. Shared Memory--A
4. Queue

36. The two ways of aborting processes and eliminating deadlocks are ____________
Answers
1. Abort all deadlocked processes

2. Abort all processes
3. Abort one process at a time until the deadlock cycle is eliminated--A
4. All of the mentioned

37. The segment limit contains the _____________
Answers
1. starting logical address of the process
2. starting physical address of the segment in memory
3. segment length--A
4. none of the mentioned

38. In the Zero capacity queue ___________
Answers
1. the queue can store at least one message
2. the sender blocks until the receiver receives the message--A
3. the sender keeps sending and the messages don't wait in the queue
4. none of the mentioned

39. What will happen if a non-recursive mutex is locked more than once?
Answers
1. Starvation
2. Deadlock--A
3. Aging
4. Signaling

40. The signal operation of the semaphore basically works on the basic ________ system call.
Answers
1. continue()
2. start()
3. wakeup()--A
4. getup()

1) What are the essential tight constraint/s related to the design metrics of an embedded system?
Select one:
a. Ability to fit on a single chip
b. Low power consumption

c. Fast data processing for real-time operations
d. All of the above

2) Mutual exclusion can be provided by the
Select one:
a. mutex locks
b. binary semaphores
c. both mutex locks and binary semaphores
d. none of the mentioned

3) Process synchronization can be done on
Select one:
a. hardware level
b. software level
c. both hardware and software level
d. none of the mentioned

4)________ techniques can be use to resolve conflicts, such as competition for resources, and
synchronize processes so that they can co-operate.
Select one:
a. Mutual Exclusion
b. Synchronization
c. Deadlock
d. Starvation

5) The page table contains
Select one:
a. base address of each page in physical memory
b. page offset
c. page size
d. none of the mentioned

6) Copying a process from memory to disk to allow space for other processes is Called
Select one:
a. Swapping
b. Deadlock
c. Demand Paging

d. Page Fault

7)Which of the following is not inode operation?
Select one:
a. link
b. unlink
c. mkdir
d. None of the above

8) Which of the following file system exports kernel objects to user space, also observes
properties of kernel internal data structures and modify them?
Select one:
a. proc
b. sysfs
c. Both
d. None

9)Which of the following architecture does not support embedded operating system
Select one:
a. ARM
b. AVR32
c. MIPS
d. None of the above

10)Select correct option ___: __: ___:___ bin:lib:sbin:etc
Select one:
a. libraries, user commands, sys admin commands, configuration files
b. user commands, sys admin commands, libraries, configuration files
c. user commands, libraries, sys admin commands, configuration files
d. user commands, libraries, configuration files, sys admin commands

12) A bootloader is responsible for i. loading an operating system kernel and its ii. loading
supporting infrastructure into memory iii. beginning the kernel's execution
Select one:
a. i and ii
b. i and iii

c. ii and iii
d. All of the above

13) Which one of the following bootloader is not used by linux?
Select one:
a. GRUB
b. LILO
c. NTLDR
d. None of the mentioned

14)Which of the following Utilities found in the binutils package
Select one:
a. as, ld, ar
b. objcopy, objdump
c. Only a
d. Only b
e. Both a and b

15. Which macro is used to add kernel module functions to kernel symbol table?
Select one:
a. MODULE_EXPORT()
b. EXPORT_SYMBOL()
c. EXPORT_KERNEL()
d. EXPORT_FUNCTION()

16) Major number identifies the driver associated with the device and Minor number is used to
identify exactly which device is referred to.
Select one:
a. True
b. False

17)Which of the following file operation in device driver allows issuing device specific commands?
Select one:
a. mmap
b. ioctl
c. write

d. None of the above

Which of the following statement is not true?
Select one:
a. insmod is used to load the kernel module.
b. device drivers are dynamically loadable modules.
c. modprobe is used to load and unload the kernel module
d. None of the above

62. Hard irq's perform non blocking and can preempted by another hard irq.
Select one or more:
a. True
b. False

Select correct option i. cdev_add() Create a device node under /dev ii. cdev_init() associates
major and minor numbers with device file iii. cdev_add() make device entry in sysfs
Select one:
a. only i is correct
b. i and ii are correct
c. i and iii are correct
d. all are correct

The first argument of printk() indicates
Select one:
a. message suffixed by priority string
b. message prefixed by priority string
c. type of kernel message logger
d. None of the above

Which of the following provides a back trace of the execution in system space with respect
to our methods?
Select one:
a. dump_stack()
b. obj_dump()
c. trace()
d. None of the above

Which of the following is external tool for debugging?
Select one:
a. probe
b. LTI
c. kgdb
d. None of the above

23) Each thread has its own user stack and no kernel stack.
Select one:
a. True
b. False

24)Which macros are used to read IO ports directly from Linux driver?
Select one:
a. read()
b. readdir()
c. inb()
d. poll()

25) Which of the is invalid name for makefile?
Select one:
a. Makefile
b. makefile
c. GNUmakefile
d. GNUMakefile
e. None of the above

26)Which of the following indicates first dependency from dependencies list?
Select one:
a. $<
b. $>
c. $@
d. $^

27)In the following command, how many jobs will run simultaneously? make -j8 -j4 bzImage
Select one:

a. 8
b. 4
c. 12
d. Both a and b
e. All of the above

28)Select correct option While configuring the kernel, the valid states for a configuration
option are: i. y — the option is enabled. ii. n — the option is not enabled. iii. m — the option
is set to be built as a module.
Select one:
a. Only i and ii
b. Only i and iii
c. i, ii and iii
d. Only ii and iii

29) Which of the following command copies the kernel image from arch/i386/boot/bzImage to
/boot/vmlinuz-version and edits /boot/grub/grub.cfg
Select one:
a. make modules
b. make modules_install
c. make bzImage
d. make install

30)"make modules" compiles all modules and "make modules_install" installs all the compiled
modules to their correct home in /modules.
Select one:
a. True
b. False

31) The entries of /sys directory
Select one:
a. are created at system startup when the subsystems register themselves with kobject core
b. are created when any device is connects with the system

c. are created at the time of kernel compilation
d. none of the mentioned

32) Which of the following is true?
Select one:
a. native compiler: A compiler where target is the same system as host.
b. cross compiler: A compiler where target is not the same system as host.
c. cross-native compiler: A native compiler where build is not the same as host.
d. All of the above.

33) To configure the kernel you can use which of the following command:
Select one:
a. make xconfig
b. make menuconfig
c. make gconfig
d. All of the above

34) Operating system can not kill any thread at any time but a thread can kill any other thread.
Select one:
a. True
b. False

35 i)make ARCH=arm menuconfig ii)make ARCH=arm CROSS_COMPILE=arm-none-linuxgnueabi
Above two lines indicates -->
Select one:
a. The first line defines the host architecture config file, and the second line defines the toolchain for host machine.
b. The first line defines the host archtecture config file, and the second line defines the cross compilation toolchain prefix
c. The first line defines the architecture that kernel needs to be built for, and the second line defines the cross compilation toolchain prefix.
d. The first line defines the architecture that kernel needs to be built for , and the second
line defines the toolchain for host machine.

36) For which of the following offset can be positive or negative?

Select one:
a. SEEK_SET
b. SEEK_END
c. SEEK_CUR
d. All of the above
e. None of the above

37) Which of the following is not unidirectional inter process communication mechanism?
Select one:
a. pipe
b. socket
c. Both
d. None

38)Which of the following signal can not be handled by the process?
Select one:
a. SIGTERM
b. SIGSTOP
c. SIGKILL
d. All of the above
e. Both a and b
f. Both b and c

39)______ option of ipcs is used show information about shared memory.
Select one:
a. -m
b. -s
c. -sm
d. None of the above

40) In which of the IPC mechanism, data is not copied from user space to kernel space and vice
a versa?
Select one:
a. Pipe
b. Message queue
c. Shared memory

d. socket

Skip to main content
SunBeam
You are logged in as E3 - Gauraw Salunke
DESD-AUG-2017 - MOCK CPT
Question 1
fork() returns non zero value in child process and zero in parent process.
Select one:
A. False
B. True
Feedback
The correct answer is: False

Question 2
Complete
Which of the following is packet based unidirectional IPC mechanism?
Select one:
A. Message queue
B. pipe
C. socket
D. None of the above
Feedback
The correct answer is: None of the above

Question 3
Which of the following command is used to delete an IPC object by id or key?
Select one:
A. ipcs
B. ipcrm
C. Ipcrm
D. ipcs -r
Feedback
The correct answer is: ipcrm

Question 4
Select correct option for mutex.
Select one:
A. A thread can lock mutex twice.
B. A thread locking mutex is owner of that mutex.
C. Owner can not unlock the mutex.
D. None of the above
Feedback

The correct answer is: A thread locking mutex is owner of that mutex.

Question 5
What is internal fragmentation?
Select one:
A. process is not utilizing the whole partition allocated to it.
B. process is utilizing the whole partition allocated to it.
C. amount of space required for process is not available.
D. amount of space required for process is available, but not contiguous.
Feedback
The correct answer is: process is not utilizing the whole partition allocated to it.

Question 6
Linux systems does not allow to create a swap file.
Select one:
A. True
B. False
Feedback
The correct answer is: False

Question 7
Physical memory : ________________ :: Logical Memory : ________________
Select one:
A. Pages, Frames
B. Frames, Pages
C. Pages, Fragments
D. fragments, Frames
Feedback
The correct answer is: Frames, Pages

Question 8
If the size of logical address space is 2 to the power of m, and a page size is 2 to the power of
n addressing units, then the high order _____ bits of a logical address designate the page
number, and the ____ low order bits designate the page offset.
Select one:
A. m, n
B. n, m
C. m – n, m
D. m – n, n
Feedback
The correct answer is: m – n, n

Question 9
LRU page replacement algorithm suffers from Belady's anomaly.
Select one:
A. true
B. false
Feedback
The correct answer is: false

Question 10
Thrashing
Select one:
A. reduces page I/O
B. decreases the degree of multiprogramming
C. implies excessive page I/O
D. improves the system performance
Feedback
The correct answer is: implies excessive page I/O

Question 11
Which of the following gives control of the CPU to the process selected by the short-term
scheduler?
Select one:
A. interpreter
B. dispatcher
C. scheduler
D. none of the mentioned
Feedback
The correct answer is: dispatcher

Question 12
While fork(), the child's set of pending signals is initially ______.
Select one:
A. filled with same as parent
B. empty
C. filled except masked signals in parent
D. None of the above
Feedback
The correct answer is: empty

Question 13
Which of the following is journaling file system
Select one:
A. JFS

B. UFS
C. ext2
D. ext3
Feedback
The correct answer is: ext3

Question 14
Which of the following utility is used to unload the kernel module?
Select one:
A. insmod
B. rmmod
C. modprobe
D. Both b and c
E. Only b
F. All of the above
Feedback
The correct answer is: Both b and c

Question 15
Which of the following file would tell you the IRQ channel for the network card while network troubleshooting?
Select one:
A. /proc/irqs
B. /proc/interrupts
C. /proc/ioports
D. /proc/interrupt
Feedback
The correct answer is: /proc/interrupts

Question 16
In kernel module programming, while reading and writing, user space address verification
can be done by_________.
Select one:
A. put_user
B. chk_user
C. access_ok
D. none of the above
Feedback
The correct answer is: access_ok

Question 17
Work Queues always works in interrupt context.
Select one:

A. True
B. False
Feedback
The correct answer is: False

Question 18
IO port allocation is done by ________
Select one:
A. check_io_region()
B. alloc_io_region()
C. register_io_region()
D. request_io_region()
Feedback
The correct answer is: alloc_io_region()

Question 19
Which of the following should be avoided in the interrupt handler?
Select one:
A. calling wait_event
B. locking semaphore
C. allocating memory
D. All of the above
Feedback
The correct answer is: All of the above

Question 20
Which of the following file stores status of the modules in linux which we can see using
lsmod?
Select one:
A. /proc/modules
B. /proc/module
C. /proc/devices
D. None of the above
Feedback
The correct answer is: /proc/modules

Question 21
The device driver module implemets ___________ function in write method.
Select one:
A. copy_to_user()
B. copy_from_user()
C. put_user()
D. get_user()

Feedback
The correct answer is: copy_from_user()

Question 22
How we can modify existing config file?
Select one:
A. make menuconfig
B. make gconfig
C. make xconfig
D. all of the above
Feedback

The correct answer is: all of the above
Question 23
The child does not inherit ________________.
Select one:
A. semaphore adjustments from its parent
B. its parent's memory locks
C. timers from its parent
D. All of the above
E. None of the above
Feedback
The correct answer is: All of the above

Question 24
Compiled kernel image is created in directory __________.
Select one:
A. /arch/x86/boot
B. /boot/grub
C. /boot/grub2
D. None of the above
Feedback
The correct answer is: /arch/x86/boot

Question 25
During execution of init_module(), state of kernel module is _______.
Select one:
A. MODULE_STATE_COMING
B. MODULE_STATE_ARRIVING
C. MODULE_STATE_GOING
D. MODULE_STATE_LIVE
E. None of the above
Feedback
The correct answer is: MODULE_STATE_COMING

Question 26
Which of the following option is used to see ELF symbol table in "objdump"?
Select one:
A. -d
B. -f
C. -t
D. -h
Feedback
The correct answer is: -t

Question 27
Which of the following datatype is not supported in module parameter?
Select one:
A. bool
B. char
C. charp
D. int
E. short
Feedback
The correct answer is: char

Question 28
Which of the following exposes module parameter with different name, outside the module?
Select one:
A. module_param()
B. module_param_name()
C. module_param_named()
D. Both b and c
E. None of the above
Feedback
The correct answer is: module_param_named()

Question 29
The device number which is of 32 bit and consist of _______ bits for major and ______ bits for
minor.
Select one:
A. 21, 11
B. 11, 21
C. 20, 12
D. 12, 20
Feedback

The correct answer is: 12, 20

Question 30
Which of the following registers cdev with kernel?
Select one:
A. cdev_init()
B. cdev_add()
C. cdev_del()
D. none of the above
Feedback
The correct answer is: cdev_add()

Question 31
Which of the following dynamically find an available device number and allocate it?
Select one:
A. register_chrdev_region()
B. alloc_chardev_regin()
C. alloc_chrdev_region()
D. All of the above
Feedback
The correct answer is: alloc_chrdev_region()

Question 32
Which of the following data structures kernel provides?
Select one:
A. Linked List
B. Queues
C. Maps
D. Binary Tree
E. All of the above
Feedback
The correct answer is: All of the above

Question 33
Which of the following is not USB driver's method
Select one:
A. probe
B. connect
C. disconnect
D. None of the above
Feedback
The correct answer is: connect

Question 34

Select odd option from below
Select one:
A. execl("./cmdline", "cmdline", "one", "two", "three", "four", NULL);
B. char *args[] = { "cmdline", "one", "two", "three", NULL }; execv("./cmdline", args);
C. execlp("ps", "ps", "-e", "-o", "pid,ppid,cmd");
D. None of the above
Feedback
The correct answer is: execlp("ps", "ps", "-e", "-o", "pid,ppid,cmd");

Question 35
Select the correct sequence for USB descriptors.
Select one:
A. device -> interface -> endpoind
B. configuration -> interface -> endpoint
C. device -> configuration -> interface -> endpoint
D. None of the above
Feedback
The correct answer is: device -> configuration -> interface -> endpoint

Question 36
In which of the following state change in child process, performing wait allows the system
to release the resources associated with child process?
Select one:
A. the child terminated
B. the child was stopped by a signal
C. the child was resumed by a signal
D. All of the above
E. None of the above
Feedback
The correct answer is: the child terminated

Question 37
Which of the following is not used to examine and change the signal action?
Select one:
A. signal
B. sigaction
C. sigprocmask
D. All of the above
Feedback
The correct answer is: sigprocmask

Question 38

Which of the following signal specified in mask, has effect on the process's signal mask
while using sigsuspend?
Select one:
A. SIGKILL
B. SIGTERM
C. SIGSTOP
D. All of the above
Feedback
The correct answer is: SIGTERM

Question 39
Select correct option for pipe
Select one:
A. pipefd[1] - read end, pipefd[0] - write end.
B. pipefd[1] - read end, pipefd[1] - write end.
C. pipefd[0] - read end, pipefd[1] - write end.
D. None of the above
Feedback
The correct answer is: pipefd[0] - read end, pipefd[1] - write end.

Question 40
Select the value of mode if O_CREAT flag is provided in open system call to give permissions
as user - read,write; group - read; others - nothing
Select one:
A. 1A0
B. 1B1
C. 1A1
D. 1B0
Feedback
The correct answer is: 1A0

Date:

Module Name: Embedded Operating Systems (DESD)

Q. No. 1

Question:

Generally, the ‘Page size’ used in Linux is

Answer Choices

A. 8 KB

B. 4 KB

C. 16 KB

D. 4 MB

Q. No. 2

Question:

The sequence of system calls that a server executes for communicating using a socket:

Answer Choices

A. socket, listen, bind, accept

B. socket, bind, listen, accept

C. socket, bind, accept, listen

D. socket, accept, bind, listen

Q. No. 3

Question:

The loader will map the ‘text’ section of an executable file to the following kind of memory segment:

Answer Choices

A. Read-Only B. Read/Write

C. Write Only D. Read/Write/Execute

Q. No. 4

Question:

The ‘read’ system call will read the data from the input stream in the following units:

Answer Choices

A. 16 bit words

B. 8 bit bytes

C. 4 bit nibbles

D. 32 bit words

Q. No. 5

Question:

The ‘pthread_join’ system call is used for

Answer Choices

A. Joining two or more threads into one

B. Sending another thread a ‘SIGJOIN’ signal

C. Terminating another thread immediately

D. Waiting for another thread to finish

Q. No. 6

Question:

We specify the IP address and the port number for the server-side socket in the following system call:

Answer Choices

A. listen

B. socket

C. bind

D. inet_aton

Q. No. 7

fork() system call will create a new child with

Answer Choices

A. A duplicate copy of all the memory segments from the parent

B. A duplicate copy of all but the stack segment from the parent

C. A fresh copy of the memory segments stuffed with zeroes

D. None of the above

Q. No. 8

Question:

In Linux, the dynamic information about a running process can be accessed from the:

Answer Choices

A. Files under the /bin directory having the names of the commands as filenames

B. Files under /dev/ directory having pids as the filenames

C. /tmp folder

D. /proc file system

Q. No. 9

Question:

Which of the following is false about the ‘malloc’ system call?

Answer Choices

A. malloc returns a void* which has to be typecast to the corresponding pointer type

B. malloc call can never fail, because it always returns some pointer

C. malloc can allocate the memory from heap segment

D. malloc call takes the argument specifying the number of bytes to be allocated

Q. No. 10

Question:

Thrashing refers to

Answer Choices

A. Kernel opening too many file descriptors

B. Stripping virtual address from physical address parts

C. System’s virtual memory subsystem going to a constant state of paging

D. A hit/miss in the Translation Look-ahead Buffer (TLB)

Q. No. 11

Question:

When a Named Pipe is created, the system does the following

Answer Choices

A. A signal gets delivered to the creating process

B. A special file gets created in the filesystem

C. The kernel executes a trap

D. The two communicating processes will enter a sleep state

Q. No. 12

Question:

UNIX pipes created with the ‘pipe’ system call provide the following type of communication:

Answer Choices

A. Multiplex B. Duplex C. Half-duplex D. Full-duplex

Q. No. 13

Question:

What is not true about System V Shared memories in UNIX?

Answer Choices

A. Shared memories can be used for IPC

B. Processes can specify unique keys to create & communicate using shared memories

C. Shared memories will always be read-only in nature

D. Shared memories have to be attached to the process' address space after creating them

Q. No. 14

Question:

The Network byte order is always

Answer Choices

A. Big Endian

B. Little Endian

C. Low Endian

D. None of the above

Q. No. 15

Question:

The ‘open’ system call returns

Answer Choices

A. A file pointer

B. A file descriptor

C. A buffer pointer

D. None of the above

Q. No. 16

Question:

Which signal number corresponds with the SIGSTOP signal?

Answer Choices

A. 5
B. 18
C. 9
D. 19

Q. No. 17

Question:

It is not the layer of the Operating system.

Answer Choices

A. Kernel
B. shell
C. Application Program
D. Critical Section

Q. No. 18

Question:

The section of code which accesses shared variables is called as

Answer Choices

A: Critical Section

B: Block

C: Semaphore

D: Procedure

Q. No. 19

Question:

The degree of Multiprogramming is controlled by

Answer Choices

A. CPU scheduler

B. Context Switching

C. Long term scheduler

D. Medium term scheduler

Q. No. 20

Question:

Process synchronization can be done on

Answer Choices

A. Hardware level

B. software level

C. both A and B

D. none of the above

Q. No. 21

Question: request_mem_region () is used to exclusively lock:

Answer Choices

A. I/O space port addresses of a device controller

B. Memory space port addresses of a device controller

C. Kernel space memory

D. User space memory

Q. No. 22

Question: request_region () is used to exclusively lock:

Answer Choices

A. I/O space port addresses of a device controller
B. Memory space port addresses of a device controller
C. Kernel space memory
D. User space memory

Q. No. 23

Question: The Major no. of a character device id in Linux 2.6 kernel is made of?

Answer Choices

A. 12 bit number
B. 20 bit number
C. 32 bit number
D. 8 bit number

Q. No. 24

Question: A kernel object file (*.ko) is different from a regular object file for the following reasons:

Answer Choices

A. Compiler used is different

B. Linker used is different

C. Kernel module related sections are added

D. All of the above

Q. No.25

Question: Which of the following errors are generated, if certain symbols are not resolved for a kernel module?

Answer Choices

A. Invalid module format

B. Unresolved symbols

C. Invalid magic no.

D. All of the above

Q. No. 26

Question: If a driver's method executing in an interrupt-context attempts an illegal memory address access, which of the following actions are taken:

Answer Choices

A. system enters a dead-lock

B. system generates an oops message

C. system generates a kernel panic

D. none of the above

Q. No. 27

Question: If a driver’s method is executing in a process context, but holding a spin-lock, what happens if it attempts to block the current process?

Answer Choices

A. system enters a dead-lock

B. system generates an oops message

C. system generates a kernel panic

D. none of the above

Q. No. 28

Question: If there is a critical section CS1 of process-context and a critical section CS2 of interrupt-context are to be exclusively executed using a lock mechanism, which lock is suitable:

Answer Choices

A. kernel mutex

B. kernel semaphore

C. kernel spin-lock

D. any of the above

Q. No. 29

Question: If timer interrupts are generated every 1ms,

mod_timer(&(dev->rx_timer), jiffies + 10); will expire after ____________

Answer Choices

A. 10ms

B. 40ms

C. 400ms

D. 4ms

Q. No. 30

Question: disable_irq() system API is typically used to:

Answer Choices

A. disable h/w interrupts on the local processor

B. disable h/w interrupts on a specific device controller

C. disable a specific h/w interrupt on the interrupt controller

D. all of the above

Q. No. 31

Question: spin_lock_irqsave() does the following:

Answer Choices

A. locks a spinlock

B. disables local processor h/w interrupts

C. disables local processor preemption

D. all of the above

Q. No. 32

Question: A typical block device file of a partition is opened, when ____________

Answer Choices

A. a file of the partition is opened

B. when the partition is mounted

C. when the partition is unmounted

D. all of the above

Q. No. 33

Question: 'inb_p' is used to ______________________

Answer Choices

A. get a byte of data from a specific port

B. get a word of data from a specific port

C. get a byte of data from a specific port with pausing

D. get a word of data from a specific port with pausing

Q. No. 34

Question: On a 32-bit CPU architecture, the size of 'jiffies' is:

Answer Choices

A. 32 bits

B. 64 bits

C. Equal to the size of 'int'

D. 16 bits

Q. No. 35

Question: Among the following, pick the one where in which semaphores can be safely used :

Answer Choices

A. Interrupt Service Routines

B. Workqueues

C. Tasklets

D. Kernel Timers

Q. No. 36

Question: You are working with Embedded Linux based hardware and you need to troubleshoot a network card problem. You want to know which IRQ channel the card is using. Which file would tell you the IRQ channel for the card?

Answer Choices

A. /proc/interrupts

B. /proc/ioports

C. /proc/irqs

D. /proc/interrupt

Q. No. 37

Question: The first stage bootloader of U-Boot (a file called MLO) in OMAP3 based architecture runs from __________?

Answer Choices

A. SRAM

B. DRAM

C. MMC card

D. All of the above

Q. No. 38

Question: A typical U-Boot boot-loader is configured and built for a specific

Answer Choices

A. embedded board

B. embedded SoC

C. embedded processor architecture

D. all of the above

Q. No. 39

Question: The kernel symbol table can be viewed by reading the file :

Answer Choices

A. /proc/ksyms

B. /proc/symbols

C. /proc/kallsyms

D. Both b and c

Q. No. 40

Question: Which of the following deferred work can sleep in the bottom half?

Answer Choices

A. Tasklets

B. Workqueues

C. Kernel Timers

D. Both a and b

Q1. Deadlocks can be avoided by
A) Acquire all resources before proceeding
B) Acquire resources in same order
C) Release resources in Reverse Order
D) All of the above.

Q2. To increase the response time and throughput, the kernel minimizes the frequency of disk access by keeping a pool of internal data buffer called
a) Pooling
b) Spooling
c) Buffer cache
d) Swapping

Q3. Which of the following enables multi-tasking in UNIX?
a) Time Sharing
b) Multi programming

c) Multi user
d) Modularity

Q4. Which of the following is considered as the super daemon in Unix?
a) sysinit
b) init
c) inetd
d) proc

Q5. On a uniprocessor system, only one process is running at a time, and it may run......
a. User mode.
b. Kernel mode.
c. Either in kernel mode or user mode.
d. Kernel mode and user mode.

Q6. Which command is used to display disk consumption of a specific directory
a) du
b) ds
c) dd
d) dds

Q7. Which of the following time stamps need not exist for a file on traditional unix file system
a) Access Time
b) Modification Time
c) Creation Time
d) Change Time

Q8. When mv f1 f2 is executed which file's inode is freed?
a) f1
b) f2
c) new inode will be used
d) implementation dependent

Q9. There are two hard links to the "file1″ say hl and h2 and a softlink sl. What happens if we deleted the "file1″?
a) We will still be able to access the file with hl and h2 but not with sl
b) We will not be able to access the file with hl and h2 but with sl
c) We will be able to access the file with any of hl, h2 and sl
d) We will not be able to access the file with any of hl, h2 and sl

Q10. Which of the following statement is true?
a) The cp command will preserve the meta data of the file

b) The sort command by default sorts in the numeric order
c) The mv command will preserve the meta data of the file
d) The command ps will display the filesystem usage

Q11. What UNIX command is used to update the modification time of a file?
a) time
b) modify
c) cat
d) touch

Q12. Which daemon manages the physical memory by moving process from physical memory to swap space when more physical memory is needed.
a) Sched daemon
b) Swap daemon
c) Init daemon
d) Process daemon

Q13. A user issues the following command sequence:
$ a.out &
$ bash
$ a.out &
If the user kills the bash process, then which of the following is true?
a) The second a.out process is also terminated
b) The second a.out process becomes a defunct process
c) The first a.out process becomes a zombie process
d) init process becomes parent of second a.out process

Q14. Which variable contains current shell process id
a) $*
b) $?
c) $$
d) $!

Q15. What is the default maximum number of processes that can exist in Linux for 32 bit system?
a) 32768
b) 1024
c) 4096
d) unlimited

Q16. How many times printf() will be executed in the below mentioned program?

```
main() {
   int i;
   for (i = 0; i < 4; i++)
   fork();

   printf("my pid = %d\n", getpid());
}
```

a) 4
b) 8
c) 16
d) 32

Q17. What is output of the following program?

```
int main() {
   fork();
   fork();
   fork();
   if (wait(0) == -1)
      printf("leaf child\n");
}
```

a) "leaf child" will be printed 1 times
b) "leaf child" will be printed 3 times
c) "leaf child" will be printed 4 times
d) "leaf child" will be printed 8 times

Q18. One process requires M resource to complete a job. What should be the minimum number of resources available for N processes so that at least one process can continue to execute without blocking/waiting?
a) M * N
b) M * N – 1
c) M * N + 1
d) M

Q19. The kill system call is used to
a) Send shutdown messages to all by super user.
b) Send a signal to a process
c) Kill processes
d) Stop the processes

Q20. Poor response time is usually caused by
a)Process busy

b)High I/O rates
c)High paging rates
d) Any of the above

Q21 The loglevel strings in printk are defined in the header file

a) <linux/module.h> b) <linux/init.h> c)<linux/kernel.h> d)<linux/fs.h>

Q22. Device that can be accessed as a stream of bytes

a)character device b)block device c)network interfaces d)none of the above

Q23. lsmod works by reading the file

a)/proc/devices b) /proc/modules c)/proc/device d)/proc/modinfo

Q24. The condition when a module fails to remove

a) kernel believes that the module is still in use.

b) the kernel has been configured to disallow module removal

c) module cannot be removed when it has dependencies.

d) All of the above

Q25. These kind of devices are not mapped to a node in filesystem

a) Serial Port b) Network c) Audio d) USB

Q26. mknod is used for creating

character device file b) block device file c) FIFO d) all of the above

Q27. Blocking read is

a)reads data continuously

b)reads data when arrives

c) blocking read operation on continuous data

d) none of these

Q28. Which mechanism is not used for delayed code execution

a) timers

b)ioctl

c) workqueues

d) tasklets

Q29. register_chrdev() does not require the argument

a) device type
b)major number
c) device name
d) device operations

Q30. Choose the incorrect statement

a) ioctl is a common interface used for device control.
b) The implementation of ioctl is mostly done using a switch statement based on the command number
c)The macro _IOR(type,nr,datatype) can be used to setup a ioctl command number with no arguments
d)The ioctl function has the format long ioctl(struct file *flip, unsigned int cmd, unsigned long arg

Q31. The driver write method implements the function

a) copy_to_user b)copy_from_user c)put_to_user d) get_to_user

Q32. Which of the following rule is false while implementing sleep

a) never sleep when you are running in an atomic context
b) there must be an event to wake up the sleeping process.
c) It is legal to sleep while holding a semaphore.

d) sleep can be done, if interrupts disabled.

Q33. Choose the most appropriate answer.

How will you identify a character device after long listing /dev directory
a)from the major number
b)from the minor number
c) by the first column of the output of longlisting.
d)by the name of the device

Q34. All the details about the module like author, version etc can be viewed by using utility

a)lsmod b)modprobe c)dmesg d)modinfo

Q35. Which file keeps the inserted driver module name with major number

a)/sys/device b) /sys/modules/ c) proc/devices d) /proc/modules

1. A process which has just terminated but has yet to relinquish its resources is called
    a. A suspended process
    b. A zombie process
    c. A blocked process
    d. A terminated process

2. Which of the following defines response time?
    a. Time from the first character of input to the first character of output.
    b. Time from the first character of input to the last character of output.
    c. Time from the last character of input to the last character of output.
    d. Time from the last character of input to the first character of output.

3. The copy-on-write mechanism provides:
    a. an efficient way to create new processes
    b. a clever way to share virtual memory pages (at least temporarily)
    c. a way to avoid unnecessary page copying

d. all of the above
e. none of the above

4. In memory management, global page replacement is usually preferable to local page replacement because:
   a. most processes are well-behaved
   b. most processes have small working sets
   c. most processes have large working sets
   d. most processes are highly synchronized
   e. the set of pages from which to choose is larger

5. Implementing LRU precisely in an OS is expensive, so practical implementations often use an approximation called:
   a. MRU
   b. MFU
   c. LFU
   d. LFU with aging
   e. none of the above

6. Counting semaphores:
   a. generalize the notion of a binary semaphore
   b. are used for managing multiple instances of a resource
   c. have increment and decrement operations
   d. can use queueing to manage waiting processes
   e. all of the above

7. The Banker's Algorithm is an example of a technique for:
   a. deadlock prevention
   b. deadlock avoidance
   c. deadlock detection
   d. deadlock recovery
   e. stabilizing turbulent financial markets

8. In message-passing IPC on a Linux system, the maximum message size permitted is
   a. 1 byte
   b. 4 bytes
   c. 32 bytes
   d. 64 bytes
   e. 140 bytes
   f. none of the above

9. When a process is created using the classical fork() system call, which of the following is not inherited by the child process?
   a. process address space

b. process ID
c. user ID
d. open files
e. signal handlers
f. none of the above

10. The threading model supported by the Linux operating system is:
   a. many-to-one
   b. one-to-one
   c. one-to-many
   d. many-to-many
   e. all of the above
   f. none of the above

11. For some file the access permissions are modified to 764.Which of the following interpretations are valid?

a. Every one can read,group can execute only and the owner can read and write

b. Every one can read and write,but owner alone can execute

c. Every one can read,group including owner can write,owner alone can execute

12. The second-chance (clock) algorithm is an efficient approximation technique for:
   a. LRU page replacement
   b. LFU page replacement
   c. benchmarking file system performance
   d. benchmarking raw disk I/O performancne
   e. two, but not all, of the above

13. When a pipe is defined--its either end can be used for reading/writing.

a. True

b. False

14. Relocatable programs
   a. cannot be used with fixed partitions
   b. can be loaded almost anywhere in memory
   c. do not need a linker

d. can be loaded only at one specific location

15. ____________ is a technique of improving the priority of process waiting in Queue for CPU allocation
    a. Starvation
    b. Aging
    c. Revocation
    d. Relocation

16. Which is not true about pipes?
    a. Pipes support broadcasting
    b. Pipes operate unidirectionally
    c. Pipes are the insecure mode of communication
    d. All the above

17. The following program: main { if (fork () > 0) sleep(1000); } results in the creation of
    a. an orphan process
    b. zombie process
    c. a process that executes forever
    d. none of the above

18. Consider the following program: main () { int i =7; if (fork() == 0) { i += 10;} else { wait(0); printf ("%d", i);}} . What will be the value of i
    a. 10
    b. 7
    c. 17
    d. Cannot be determined

19. Which of the following system calls, does not return control to the calling point, on termination?
    a. Fork
    b. exec
    c. wait
    d. longjmp

20. Situations where two or more process are reading or writing some shared data and the final result depends on the order of usage of the shared data are called
    a. Race Conditions
    b. Critical-Section
    c. Mutual Exclusion
    d. Deadlocks

21. Preemption is
   a. forced de allocation of the CPU from a program which is executing on the CPU.
   b. release of CPU by the program after completing its task.
   c. forced allotment of CPU by a program to itself.
   d. a program terminating itself due to detection of an error

22. Using Priority Scheduling algorithm, find the average waiting time for the following set of processes given with their priorities in the order:

Process	Burst Time	Priority respectively
P1	10	3 ,
P2	1	1 ,
P3	2	4 ,
P4	1	5 ,
P5	5	2.

   a. 8 milliseconds
   b. 8.2 milliseconds
   c. 7.75 milliseconds
   d. 3 milliseconds

23. Go through the following C program

```
main ()
{int i, n ;
for (i = 1 ; i <= n ; ++i) fork () ;
printf("yes") ;  }
```
For what value of n, will yes be printed 24 times?
   a. 3
   b. 4
   c. 5
   d. Impossible to find such an n

24. Consider the following program
```
main ( ) {
 int p[2] ;pipe (p); fork () ; }
```
   Choose the correct answers.
   a. The pipe will be recognized only by the parent process
   b. p [0] is the file descriptor of the write end of the pipe
   c. There will be 4 file descriptors in memory
   d. The pipe will be shared by both the parent and the child processes

25. High Paging Activity leads to
   a. Swapping
   b. Compaction
   c. Thrashing

d. External Fragmentation

26. The call pipe (p); is valid if p had been declared as
    a. int p
    b. int p[2]
    c. char *p
    d. FILE *p

27. What is a shell ?
    a. command interpreter
    b. hardware component
    c. part of compiler
    d. tool used in scheduling

28. The mechanism that brings page into memory only when it is needed is called
    a. Segmentation
    b. Fragmentation
    c. Page Replacement
    d. Demand Paging

29. Virtual memory is ___________
    a. An extremely large main memory
    b. An extremely large secondary memory
    c. An illusion of extremely large main memory
    d. A type of memory used in super computers

30. The process related to process control, file management, device management, information about system and communication that is requested by any higher level language can be performed by ___________
    a. Editors
    b. Compilers
    c. System Call
    d. Caching

31. Information about a process is maintained in a _________
    a. Stack
    b. Translation Look aside Buffer
    c. Process Control Block
    d. Program Control Block

32. Poor response time is usually caused by
    a. Process busy
    b. High I/O rates
    c. High paging rates
    d. Any of the above

33. A system program that combines the separately compiled modules of a program into a form suitable for execution
    a. assembler

b. linking loader
c. cross compiler
d. load and go
e. None of the above

34. Which signal is generated when we press ctrl-Z?

A) SIGKILL

B) SIGTSTP

C) SIGABRT

D) SIGINT

35. Thrashing
a. is a natural consequence of virtual memory systems
b. can always be avoided by swapping
c. always occurs on large computers
d. can be caused by poor paging algorithms
e. None of the above

36. The number of process completed per unit time is known as____

A) Output

B) Throughput

C) Efficiency

D) Capacity

37. Fragmentation of the file system
a. occurs only if the file system is used improperly
b. can always be prevented
c. can be temporarily removed by compaction
d. is a characteristic of all file systems
e. None of the above

38. The initial value of the semaphore that allows only one of the many processes to enter their critical sections, is
a. 8
b. 1
c. 16
d. 0
e. None of the above

39. Which of the following statements is true about Shared Memory?

A)Shared Memory have in-built synchronization

B)Shared Memory are efficient for variable size data transfers

C)Shared Memory are faster as compared to Message Queues

D)Shared Memory is maintained in the kernel address space

40. In Priority Scheduling a priority number (integer) is associated with each process. The CPU is allocated to the process with the highest priority (smallest integer = highest priority). The problem of, Starvation ? low priority processes may never execute, is resolved by ___________.
    a. Terminating the process
    b. Aging
    c. Mutual Exclusion
    d. Semaphore

1. The component responsible for moving pages in and out of physical memory is

Long term scheduler

b) Short term scheduler

Medium term scheduler

d) Dispatcher

2. Banker's algorithm is used for?

Deadlock Avoidance

Deadlock Prevention

Deadlock Recovery

d. Deadlock Detection

3. Which of the following CPU scheduling algorithm will prevent starvation problem?

Shortest job first b) Priority scheduling

c) Priority scheduling with aging mechanism d) none of the above

4. What does the CPU do when there are no programs to run?

a) Sleeps b) Runs a busy wait loop

c) Shuts down the system d) none of the above

5. When a computer is first turned on or restarted, a special type of absolute loader called ____ is executed.

Compile and Go loader

Boot loader

Bootstrap loader

Relating loader

6. The operating system manages

Memory

Processes

Disks and I/O devices

All of the above

7. Swapping

Works best with many small partitions

Allows many programs to use memory simultaneously

Allows each program in turn to use the memory

Does not work with overlaying

8. Critical section concurrently accessed by

More than one thread of execution

Two thread of execution

One thread of execution

All of the above

9. Which of the following is not an advantage of multiprogramming?

a) Increase through put   b) Shorter Response Time

c) Decrease operating system Overhead   d) Ability to design priorities to jobs

10. Multiprocessing models have

I. Symmetric Multiprocessing model   II. Loosely coupled system

III. Tightly coupled system   IV. Parallel system

a) I, II & IV only   b) I & III only

c) I & II only   d) I, III & IV only

11. Timesharing is the same as

a) multitasking b) multiprogramming c) multiuser d) none of these

12. Which of the following instructions can be executed in kernel mode?

Reboot

Disable Interrupts

Write the instruction register

None of these

13. Which of transition between process states is not possible?

a. Ready (Run

b. Ready (held

c. Blocked (Ready

d. Swapped-blocked (swapped-ready

14. Which one controls “degree of multiprogramming?”

a. Job scheduler

b. CPU scheduler

c. Both a &b

d. None of these

15. Which of the following are preemptive scheduling scheme?

Shortest job first

Shortest remaining time

First come first serve

None of These

16. Which of the following is page replacement policy using this strategy" Replace the page that hasn't been used for the longest period of time"?

Least recently used

Not recently Used

Clock

Optimal Page Replacement

17. What is the effect of fragmentation?

a. Increased complexity of algorithm

b. Leads to wastage of memory

Leads to page faults

Helps in multitasking

18. Which of the following is not an advantage of multiprogramming?

a) Increase through put　　b) Shorter Response Time

c) Decrease operating system　　d) Ability to design priorities to jobs

Overhead

19. time-stamping is

a. used to order events in a distributed system without the use of clocks.

b. used to order events in a distributed system with the use of clocks.

c. used to order events in a time shared system without the use of clocks

d. used to order events in a time shared system with the use of clocks

20. What is swapping?

Copying a process from memory to disk to allow space for other processes

Copying a job in disk from place to other place in same disk

Copying a process from memory to cache

None of the above

21. Calculate the average waiting time of given processes, applying the priority based primitive Scheduling and also make Gantt chart. (Assume Lower number having higher priority)

5 Marks

Process Priority Burst Time Arrival Time

P1	2	4	2
P2	1	6	4
P3	3	7	0
P4	4	5	1
P5	2	6	3

Ans………………………

Linux Internal

1. Program execution is mainly takes place in

a) user mode b) kernel mode

c) Both d) none of these

2-A switch to kernel node can be triggered by

a) System call b) exception

c) interrupt d) all of the above

3-When kernel moves in kernel mode

it running on behalf of kernel Process

b) it running on behalf of user process

it running on behalf of kernel and user process

d) none of these

4-System calls for IPC are

a) pipe()

b) dup()

c) fork()

d) none of the above

5-What will be output

```
main()
{
        cout<<”hello world”;
    fork();
    fork();
        cout<<”hello world”;
    fork();
    fork();
}
```

a) 16 times

b) 8 times

c) 4 times

d) none of these

6-It is possible that a file descriptor would be in negative number

a) yes b) no

c) in some case it is possible d) none of these

7-Normally kernel mode is represented by

a) 1 b) 2

c) 0 d) none of these

8. PROCESS ID(PID) = 0 IS STAND FOR

a)Sawapper

b)init

c)Dispacher

d)All the above

9. A process P1 creates a child process P2 . P1 terminates. What is the status of P2

a)P2 hangs

b)P2 is run to completion only by the init.

c)P2 becomes a zombie process

d)P2 is run to completion by any of the parent of P1

10. kill (pid_t pid, int sig); return

a) Zero if successful

b) Id of parent process

c) Id of child process

d) Return the process identifier of current process.

11. main()

```
{
int pid;
pid=fork();
if(pid==0)
{
sleep(2);
printf("%d %d \n", getpid(),getppid());
}}
```

if pid of child process=245;

if pid of parent process=242;

what will be the o/p;

a)245 242 b)242 1

c)245 1 d)242 245

12. main ()

```
        {
int pid;
printf(“hi\n”);
pid=fork();
printf(“Hello!\n”);
if(pid==0)
{
        printf (“hi”\n);
        printf (“hi\n”);
        printf ("Hello !\n");
        }
        printf(“Hello!\n”);
        }
```

a) “hi” three times and “Hello!” Six times

b) “hi” three times and “Hello!” five times

c) “hi” five times and “Hello!” Six times

d) “hi” three times and “Hello” three times

13. Signal number for SIGUSR1 is

a. 2

b. 9

c. 10

d. 11

14. In message queue message type is always

string

b. char

c. long

d. short

e. unsigned int

15. Socket is an interface between an application process and ……………

a. Transport Layer

b. Another Host system

c. Network layer

d. Application Layer

16. What is the value of semaphore variable in case of counting semaphore?

1

Total number of resources.

Total number of available resources.

Total number of processes.

17. Process opens an existing file or creates a new file; kernel returns a non-negative integer, what is the return integer?

a) Inode number

b) Process Indentifier

c) File descriptor

d) Process return value

18. What is the use of SIGCHLD signal?

Kernel notifies the parent process when process terminates.

When child process becomes orphan process.

Creation of child process with fork

None of these.

19. WIFEXITED (status) macro returns true when

Child terminated abnormally, by receipt of a signal that it didn’t catch.

Child that terminated normally.

Child that is currently stopped.

None of these

20. What is the Signal Number for SIGUSR1?

a.) 4 b.)6

c.) 8 d.)10

1. The component responsible for moving pages in and out of physical memory is

a) Long term scheduler b) Short term scheduler
c) Medium term scheduler d) Dispatcher

2. A system has 3 processes sharing 4 resources. If each process needs maximum 2 then

a) Deadlock can never occur b) Deadlock may occur
c) Deadlock has to occur d) none of the above

3. Which of the following CPU scheduling algo will prevent starvation problem?

a) Shortest job first
b) Priority scheduling
c) Priority scheduling with aging mechanism
d) none of the above

4. What does the CPU do when there are no programs to run?

a) Sleeps
b) Runs a busy wait loop
c) Shuts down the system
d) none of the above

5. When a computer is first turned on or restarted, a special type of absolute loader called ____ is executed

a) Compile and Go loader
b) Boot loader
c) Bootstrap loader
d) Relating loader

6. The operating system manages

a) Memory
b) Processes
c) Disks and I/O devices
d) All of the above

7. What is the name given to the organized collection of software that controls the overall operation of a computer?

a. Working system
b. Peripheral system
c. Operating system
d. Controlling System

8. Swapping

a) Works best with many small partitions
b) Allows many programs to use memory simultaneously
c) Allows each program in turn to use the memory
d) Does not work with overlaying

9. Critical section concurrently accessed by

a. More than one thread of execution
b. Two thread of execution
c. One thread of execution
d. All of the above

10. Which of the following is not an advantage of multiprogramming?

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b) I & III only
c) I & II only
d) I, III & IV only

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13. Which of the following instructions can be executed in kernel mode?

a. Reboot
b. Disable Interrupts
c. Write the instruction register
d. None of these

14. Which of transition between process states is not possible?

e. Ready → Run
f. Ready → held
g. Blocked → Ready
h. Swapped-blocked → swapped-ready

15. Which one controls "*degree of multiprogramming?*"

Job scheduler
CPU scheduler
Both a &b
None of these

Q1. Program execution is mainly takes place in

a) user mode b) kernel mode

c) Both d) none of these

Q2-A switch to kernel node can be triggered by

a) System call b) exception

c) interrupt d) all of the above

Q3-When kernel moves in kernel mode

a) it running on behalf of kernel Process b) it running on behalf of user process

b) it running on behalf of kernel and user process d) none of these

Q4-System calls for IPC are

a) pipe() b) dup()

c) fork() d) none of the above

Q5-What will be output

```
main()
{
        cout<<"hello world";
    fork();
    fork();
        cout<<"hello world";
    fork();
```

```
        fork();
    }
```

a) 16 times b) 8 times

c) 4 times d) none of these

Q6-It is possible that a file descriptor would be in negative number

a) yes b) no

c) in some case it is possible d) none of these

Q7-Normally kernel mode is represented by

a) 1 b) 2

c) 0 d) none of these

Q 8. PROCESS ID(PID) = 0 IS STAND FOR

a)Sawapper

b)init

c)Dispacher

d)All the above

Q9. A process P1 creates a child process P2 . P1 terminates. What is the status of P2

a)P2 hangs

b)P2 is run to completion only by the init.

c)P2 becomes a zombie process

d)P2 is run to completion by any of the parent of P1

Q10. kill (pid_t pid, int sig); return

a) Zero if successful

b) Id of parent process

c) Id of child process

d) Return the process identifier of current process.

Q11. main()

```
{
int pid;
pid=fork();
if(pid==0)
{
sleep(2);
printf("%d %d \n", getpid(),getppid());
}}
```

if pid of child process=245;

if pid of parent process=242;

what will be the o/p;

a)245 242   b)242 1

c)245 1   d)242 245

Q12. main ()

```
{
int pid;
```

```
printf("hi\n");
pid=fork();
printf("Hello!\n");
if(pid==0)
{
printf ("hi"\n);
printf ("hi\n");
printf ("Hello !\n");
}
printf("Hello!\n");
}
```

a) “hi” three times and “Hello!” Six times

b) “hi” three times and “Hello!” five times

c) “hi” five times and “Hello!” Six times

d) “hi” three times and “Hello” three times

13. Signal number for SIGUSR1 is

a. 2 b. 9
c. 10 d. 11

14.Which Signal handler ignores the generated Signal?

a. SIG_DFL b. SIG_IGNR
b. SIGSEGV d. None of these

15.In message queue message type is always

a. unsigned int b. long

c. char d. short

1. Modern GPOS operating systems are built, using : 0/1
A) only core components
B) only non-core components
C) core and non-core components
D) none of the above

2. In modern GPOS operating systems, which of the following is True: 0/1
A) core-components are isolated, using memory protection
B) core-components are not isolated, using memory protection
C) core-components and non-core components reside, in a single address-space
D) none of the above

3. system call APIs of a GPOS are used to access : 0/1
A) process management services only
B) non-core services only
C) networking services only
D) all core services

4. system call system routines of system call APIs are resident, in :0/1
A) system libraries B) user-space C) system-space D) none of the above

5. An appropriate user-space representation of a process is :0/1
A) process address-space of the process instance
B) a process descriptor of the process instance
C) Input/Output resources allocated to the process
D) active files of the process instance

6. An appropriate system-space representation of a process is :0/1
A) process address-space of the process instance
B) process descriptor of the process instance
C) program associated with the process instance
D) scheduling parameters associated, with the process instance

7. In a modern GPOS system :1/1
A) processes support multitasking, in uni-processor systems only
B) processes support multitasking, in multi-processor systems only
C) processes support multitasking, in uni-processor and multi-processorsystems
D) none of the above

8. In a modern GPOS system, applications' processes are assigned :1/1
A) +ve values, for process ids
B) -ve values, for process ids
C) +ve and -ve values, for process ids
D) none of the above

9. By default, time-sharing scheduler, in a modern GPOS assigns :1/1
A) equal time-shares to processes
B) unequal time-shares to processes
C) time-shares, based on the user-id of the process descriptor
D) time-shares, based on the application program's code size

10. in a modern GPOS, default scheduling policy is :1/1
A) real-time priority scheduling
B) real-time round-robin scheduling
C) time-sharing scheduling
D) none of the above

11. message-queue IPC mechanism, in a modern GPOS is :0/1

A) an asynchronous mechanism and supports synchronization implicitly

B) an asynchronous mechanism, but does not support synchronization implicitly

C) a synchronous mechanism and supports synchronization implicitly

D) a synchronous mechanism, but does not support synchronization implicitly

12. a sempahore, in a modern GPOS supports :1/

A) locking and unlocking operations, with blocking suppor

B) decrement and increment operations, with blocking support

C) locking and unlocking operations, without blocking support

D) decrement and increment operations, without blocking support

13. decrement and increment operations of a semaphore areimplemented, in a modern GPOS, as1/1

A) atomic operations, with blocking support

B) non-atomic operations, with blocking support

C) atomic operations, without blocking support

D) non-atomic operations, without blocking support

14. Understand the following concurrent applicationand select thecorrect answer.0/1

A) there is a single process and multiple threads

B) there is are multiple processes and multiple threads

C) there is a single parent process and multiple children processes

D) none of the above

15. For a shared-memory IPC mechanism, which of the following is set-up:1/1

A) a special private virtual-memory segment is set-up, in process address-space

B) a special shared virtual-memory segment is set-up, in process address-space

C) a special private virtual-memory segment is set-up, in kernel address-space

D) a special shared virtual-memory segment is set-up, in kernel address-space

16. Each process, in a modern GPOS is allocated, during process creation:1/1
A) a shared virtual address-space, without protection
B) a private virtual address-space, without memory protection
C) a private virtual address-space, with memory protection
D) none of the above

17. data-race conditions can occur, in which of the following scenarios :1/1
A) if there is no shared-data used, in concurrent operations
B) if there is shared-data used, in non-concurrent operations
C) if there is shared-data used, in concurrent operations
D) none of the above

18. Maximum size allowed, for an active application/process is :0/1
A) very large, in the order of tera-bytes, in a 32-bit hw system
B) very large, in the order of tera-bytes, in a 64-bit hw system
C) large, in the order of giga bytes, in a 64-bit system
D) none of the above

19. which of the following components of a GPOS reside, in system-space :1/1
A) thread libraries
B) device drivers
C) system utilities
D) system call library

20. PATH environment variable is used to find :0/1
A) external commands only
B) internal commands of the shell only
C) external commands and internal commands of the shell
D) none of the above

21. if a system utility is executed, in a super-user shell :0/1

A) the system utility is executed, in a new child process, with super-user credentials

B) the system utility is executed, in the shell, with super-user credentials

C) the system utility is executed, in a new child process, with higher processorprivileges

D) the system utility is executed, in a shell, with higher processor privileges

22. if a Linux process terminates, it is immediately moved to :0/

A) STOPPED state

B) DEAD state

C) ZOMBIE state

D) UNINTERRUPTIBLE blocked state

23. in a well behaved Linux GPOS system, swap-space is used, for :0/1

A) storing less active virtual pages of processes

B) storing active virtual pages of processes

C) storing less active and active pages of processes

D) storing less active kernel-space virtual pages

24. waitpid() system call API is used to : 0/1

A) terminate a child process

B) clean-up a terminated child process

C) terminate and clean-up a child process

D) none of the above

25. if we execute renice -n 10 -p <pid>, as a normal user : 0/1

A) the command is successfully executed

B) the command generates a processor exception

C) the command is unsuccessful

D) none of the above

Operating System MCQ's

What is operating system? a) collection of programs that manages hardware resources b) system service provider to the application programs c) link to interface the hardware and application programs d) all of the mentioned
Ans : d
Dual mode of operating system has A)1Mode (B)2Modes (C) 3 Modes (D)4 Modes
Ans : B
To access the services of operating system, the interface is provided by the a) System calls (b) API (c) Library (d) Assembly instructions
Ans:A
Which one of the following error will be handled by the operating system? a) power failure b) lack of paper in printer c) connection failure in the network d) all of the mentioned
Ans:D
The main function of the command interpreter is a) to get and execute the next user-specified command b) to provide the interface between the API and application program c) to handle the files in operating system d) none of the mentioned
Ans:A

The systems which allows only one process execution at a time, are called a) uniprogramming systems b) uniprocessing systems c) unitasking systems d) none of the mentioned
Ans:B

Environment in which programs of the computer system are executed is: (a)OS (b)Nodes (c)Clustered System (d)both a and b
And : A
A properly designed operating system must ensure that an incorrect (or malicious) program cannot cause other programs to execute (a)Incorrectly (b)Correctly (c) Both a and b (d)None
Ans: A
The user view of the system depends upon the (a)CPU (b)Software (c)Hardware (d)Interface
Ans:D
Control and Status registers are used by processor to control A. Design of the Processor B. Operation of the Processor C. Speed of the Processor D. Execution of the Processor
Ans: b
Program execution services are used to A. Control Program B. Delete Program C. Execute Program D. Update Programs
Ans :C
Readfile() call function in windows operating system is a UNIX's function called for A. fork() B. open() C. read() D. close()
Ans: C
Bootstrap program that starts operating system is normally stored in A. RAM

B. ROM C. hard disk D. CD
Ans:B
Kernel mode of operating system runs when mode bit is A. 1 B. 0 C. x D. undefined
Ans:B
Whenever the data is found in the cache memory it is called as __________ a) HIT b) MISS c) FOUND d) ERROR
Ans:A
The transfer between CPU and Cache is ________________ a) Block transfer b) Word transfer c) Set transfer d) Associative transfer
Answer:b
A system call is a routine built into the kernel and performs a basic function. a) True b) False Ans: A

The chmod command invokes the_____system call. a) chmod b) ch c) read d) change Ans: A
Which of the following system call is used for opening or creating a file? a) read b) write c) open d) close Ans:C
I/O modules performs requested action on A. Programmed I/O B. Direct Memory Access (DMA) C. Interrupt driven I/O D. I/O devices Ans:A
Kernel mode of operating system runs when the mode bit is (a)1 (b)0 (c)X (d)undefined Ans:B
Addresses of interrupt programs of operating system are placed at A. Interrupt cell routine B. Interrupt call service C. interrupt vector table

D. interrupt service routine

Ans: C

Which scheduling algorithm allocates the CPU first to the process that requests the CPU first?
a) first-come, first-served scheduling
b) shortest job scheduling
c) priority scheduling
d) none of the mentioned

Ans:A

In multilevel feedback scheduling algorithm
a) a process can move to a different classified ready queue
b) classification of ready queue is permanent
c) processes are not classified into groups
d) none of the mentioned

Ans:A

The two steps of a process execution are :
a) I/O & OS Burst
b) CPU & I/O Burst
c) Memory & I/O Burst
d) OS & Memory Burst

Ans:B

Scheduling is done so as to :
a) increase CPU utilization
b) decrease CPU utilization
c) keep the CPU more idle
d) None of the mentioned

Ans:A

Round robin scheduling falls under the category of :
a) Non preemptive scheduling

b) Preemptive scheduling c) All of the mentioned d) None of the mentioned   Ans:B
Scheduling of threads are done by A. input B. output C. operating system D. memory Ans:C
‘Aging’ is : a.keeping track of cache contents b.keeping track of what pages are currently residing in memory c.keeping track of how many times a given page is referenced d.increasing the priority of jobs to ensure termination in a finite time   Ans:D
To overcome the slow operating speeds of the secondary memory we make use of faster flash drives. a) True b) False Ans:A
The next level of memory hierarchy after the L2 cache is a) Secondary storage b) TLB c) Main memory d) Register

Ans:c

Which algorithm suffers from Convoy Effect?
a.FCFS
b)SJF – Non preemptive
. c)SJF – Preemptive
d.Round Robin

Ans:A

An IPC facility provides at least two operations :
a) write & delete message
b) delete & receive message
c) send & delete message
d) receive & send message
Ans:D

The segment of code in which the process may change common variables, update tables, write into files is known as :
a) program
b) critical section
c) non – critical section
d) synchronizing
Ans:B

To enforce ………………….. two functions are provided enter-critical and exit-critical, where each function takes as an argument the name of the resource that is the subject of competition.
A) Mutual Exclusion
B) Synchronization
C) Deadlock
D) Starvation
Ans:A

A system is in the safe state if
a) the system can allocate resources to each process in some order and still avoid a deadlock

b) there exist a safe sequence c) all of the mentioned d) none of the mentioned Ans:A
While preventing deadlock with .... needs no run-time computation since problem is solved in system design. A) request all resources B) preemption C) resource ordering D) finding safe path Ans:C
Physical memory is broken into fixed-sized blocks called _________ a) frames b) pages c) backing store d) none of the mentioned Ans:A
Every address generated by the CPU is divided into two parts : a) frame bit & page number b) page number & page offset c) page offset & frame bit d) frame offset & page offset Ans:B
The______table contains the base address of each page in physical memory. a) process b) memory c) page d) frame

Ans:C
Virtual memory is normally implemented by _________ a) demand paging b) buses c) virtualization d) all of the mentioned Ans:A
The valid – invalid bit, in this case, when valid indicates : a) the page is not legal b) the page is illegal c) the page is in memory d) the page is not in memory Ans:C
The three major methods of allocating disk space that are in wide use are : a) contiguous b) linked c) indexed d) all of the mentioned Ans:D
In indexed allocation : a) each file must occupy a set of contiguous blocks on the disk b) each file is a linked list of disk blocks c) all the pointers to scattered blocks are placed together in one location d) none of the mentioned Ans:C
The major disadvantage with linked allocation is that : a) internal fragmentation b) external fragmentation c) there is no sequential access d) there is only sequential access Ans:D
Indexed allocation___________direct access.

a) supports b) does not support c) is not related to d) none of the mentioned Ans:A
In the________algorithm, the disk arm starts at one end of the disk and moves toward the other end, servicing requests till the other end of the disk. At the other end, the direction is reversed and servicing continues. a) LOOK b) SCAN c) C-SCAN d) C-LOOK Ans:B
In the_________algorithm, the disk head moves from one end to the other , servicing requests along the way. When the head reaches the other end, it immediately returns to the beginning of the disk without servicing any requests on the return trip. a) LOOK b) SCAN c) C-SCAN d) C-LOOK Ans:C

Operating System MCQ's Set1

Question No.	Questions & Answers
1	What is operating system? a) collection of programs that manages hardware resources b) system service provider to the application programs c) link to interface the hardware and application programs d) all of the mentioned
	Ans : d
2	Q2.Dual mode of operating system has A)1Mode (B)2Modes (C) 3 Modes (D)4 Modes
	Ans : B

3	Q3. 2. To access the services of operating system, the interface is provided by the a) System calls (b) API (c) Library (d) Assembly instructions
	Ans:A
4	Which one of the following is not true? a) kernel is the program that constitutes the central core of the operating system b) kernel is the first part of operating system to load into memory during booting c) kernel is made of various modules which can not be loaded in running operating system d) kernel remains in the memory during the entire computer session
	Ans:C
5	Which one of the following error will be handle by the operating system? a) power failure b) lack of paper in printer c) connection failure in the network d) all of the mentioned
	Ans:D

6	The main function of the command interpreter is a) to get and execute the next user-specified command b) to provide the interface between the API and application program c) to handle the files in operating system d) none of the mentioned
	Ans:A
7	The systems which allows only one process execution at a time, are called a) uniprogramming systems b) uniprocessing systems c) unitasking systems d) none of the mentioned
	Ans:B
8	Example of open source operating system is (a)Unix (b)Linux (c) Windows (d)both a & b
	Ans : D
9	Environment in which programs of the computer system are executed is: (a)OS (b)Nodes (c)Clustered System (d)both a and b
	And : A
10	The main function of the command interpreter is: A. to get and execute the next user-specified command B. to provide the interface between the API and application program C. to handle the files in operating system D. none of the mentioned
	Ans:A
11	By operating system, the resource management can be done via: • A. time division multiplexing • B. space division multiplexing • C. both (a) and (b)

	• D. none of the mentioned
	Ans : C
12	If a process fails, most operating system write the error information to a: • A. log file • B. another running process • C. new file • D. none of the mentioned
	Ans:A
13	A properly designed operating system must ensure that an incorrect (or malicious) program cannot cause other programs to execute (a)Incorrectly (b)Correctly (c) Both a and b (d)None
	Ans: A
14	The user view of the system depends upon the (a)CPU (b)Software (c)Hardware (d)Interface
	Ans:D
15	Control and Status registers are used by processor to control A. Design of the Processor B. Operation of the Processor C. Speed of the Processor D. Execution of the Processor
	Ans: b
16	Kernel mode of the operating system is also called (a) User mode (b)system mode (c)supervisor mode (d)both a and b
	Ans:C
17	Error detection and response clears the A. Program B. Data C. Information D. Error Condition
	Ans:D

18	Program execution services are used to A. Control Program B. Delete Program C. Execute Program D. Update Programs
	Ans :C
19	Access control in operating system is just another name for A. Data manipulation B. Files Access C. Compartmentalization of resources D. Data and Resources Access
	Ans:C
20	Operating system provides System access function to protect A. I/O Modules B. Computer C. Memory D. Data and Resources
	Ans:D
21	Readfile() call function in windows operating system is a UNIX's function called for A. fork() B. open() C. read() D. close()
	Ans: C
22	The kernel is_________of user threads. a) a part of b) the creator of c) unaware of d) aware of
	Ans: C
23	Because the kernel thread management is done by the

	Operating System itself : a) kernel threads are faster to create than user threads b) kernel threads are slower to create than user threads c) kernel threads are easier to manage as well as create then user threads d) none of the mentioned	
	Ans:b	
24	Kernel mode of operating system is also called A. user mode B. system mode C. supervisor mode D. Data mode	
	Ans:C	
25	Which of the following are the functions of operating system? i) recovering from errors ii) facilitating input/output iii) facilitating parallel operation iv) sharing hardware among users v) implementing user interface a. I,ii,and v only b.i,ii,iii and iv only c. ii,iii,iv and v only d.i,ii,iii,iv and v	
	Ans : D	
26	1kilobyte memory storage in form of bytes is equal to A. 1024 bytes B. 1026 bytes C. 1056 bytes D. 1058 bytes	
	Ans :A	
27	Bootstrap program that starts operating system is normally stored in A. RAM	

	B. ROM C. hard disk D. CD
	Ans:B
28	Interrupts which are initiated by an instruction are (a)Internal(B)External(C)Hardware (D)Software
	Ans. D
29	Example of open source operating system is A. UNIX B. Linux C. windows D. both a and b
	Ans: D
30	Kernel mode of operating system runs when mode bit is A. 1 B. 0 C. x D. undefined
	Ans:B
31	To access the services of operating system, the interface is provided by the A. system calls B. API C. library D. assembly instructions
	ANSWER: A
32	Cache memory is used A. to avoid speed mismatch B. to storage the data C. for data accusation D. none of the above
	ANSWER: A
33	What is the high speed memory between the main memory

	and the CPU called? a) Register Memory b) Cache Memory c) Storage Memory d) Virtual Memory
	Ans:B
34	Cache Memory is implemented using the DRAM chips. a) True b) False
	Answer: b Explanation: The Cache memory is implemented using the SRAM chips and not the DRAM chips. SRAM stands for Static RAM. It is faster and is expensive.
35	Whenever the data is found in the cache memory it is called as a) HIT b) MISS c) FOUND d) ERROR
	Ans:A
36	When the data at a location in cache is different from the data located in the main memory, the cache is called _______________ a) Unique b) Inconsistent c) Variable d) Fault
	Ans:B
37	The transfer between CPU and Cache is a) Block transfer b) Word transfer c) Set transfer d) Associative transfer
	Answer:b

	Explanation: The transfer is a word transfer. In the memory subsystem, word is transferred over the memory data bus and it typically has a width of a word.
38	Levels between CPU and main memory were given a name of A.Hit time B.Miss Rate C.Locality in time D.Cache
	Ans.D

Operating System MCQ Day2

1	A system call is a routine built into the kernel and performs a basic function. a) True b) False Ans: A

2	When we execute a C program, CPU runs in_____mode. a) user b) kernel c) supervisory d) system Answer: a Explanation: When we execute a C program, the CPU runs in user mode. It remains it this particular mode until a system call is invoked.
3	In___________mode, the kernel runs on behalf of the user.

	a) user b) kernel c) real d) all Ans:b Explanation: Whenever a process invokes a system call, the CPU switches from user mode to kernel mode which is a more privileged mode. The kernel mode is also called as supervisor mode. In this mode, the kernel runs on behalf of the user and has access to any memory location and can execute any machine instruction.
4	All UNIX and LINUX systems have one thing in common which is a) set of system calls b) set of commands c) set of ________ instructions d) set of text editors

	Ans:A
5	The chmod command invokes the ________ system call. a) chmod b) ch c) read d) change Ans: A Explanation: Many commands and system calls share the same names.
6	For reading input, which of the following system call is used? a) write b) rd c) read d) change Ans:C
7	Which of the following system call is

	used for opening or creating a file? a) read b) write c) open d) close Ans:C
8	System call routines of operating system are mostly written in A. C B. C++ C. java D. both a and b Ans:D
9	I/O modules performs requested action on A. Programmed I/O B. Direct Memory Access (DMA) C. Interrupt driven I/O

	D. I/O devices Ans:A
10	Control and Status registers are used by processor to control A. Design of the Processor B. Operation of the Processor C. Speed of the Processor D. Execution of the Processor Ans:B
11	Kernel mode of operating system runs when the mode bit is (a)1 (b)0 (c)X (d)undefined Ans:B
12	One that is not a type of memory is A. cache B. ROM C. RAM

	D. compilers Ans:D
13	I/O instruction transfer is used to read the A. Data B. Information C. Instructions D. Description Ans:A
14	Addresses of interrupt programs of operating system are placed at A. Interrupt cell routine B. Interrupt call service C. interrupt vector table D. interrupt service routine Ans: C
15	Which module gives control of the CPU to the process selected by the short-term scheduler?

	a) dispatcher b) interrupt c) scheduler d) none of the mentioned Ans:A
16	The processes that are residing in main memory and are ready and waiting to execute are kept on a list called a) job queue b) ready queue c) execution queue d) process queue Ans: B
17	The interval from the time of submission of a process to the time of completion is termed as a) waiting time b) turnaround time c) response time

	d) throughput Ans:B
18	Which scheduling algorithm allocates the CPU first to the process that requests the CPU first? a) first-come, first-served scheduling b) shortest job scheduling c) priority scheduling d) none of the mentioned Ans:A
19	In priority scheduling algorithm a) CPU is allocated to the process with highest priority b) CPU is allocated to the process with lowest priority c) Equal priority processes can not be scheduled d) None of the mentioned

	Ans:A
20	In priority scheduling algorithm, when a process arrives at the ready queue, its priority is compared with the priority of a) all process b) currently running process c) parent process d) init process Ans:B
21	Time quantum is defined in a) shortest job scheduling algorithm b) round robin scheduling algorithm c) priority scheduling algorithm d) multilevel queue scheduling algorithm Ans:B
22	Process are classified into different groups in a) shortest job scheduling algorithm

	b) round robin scheduling algorithm c) priority scheduling algorithm d) multilevel queue scheduling algorithm Ans:D
23	In multilevel feedback scheduling algorithm a) a process can move to a different classified ready queue b) classification of ready queue is permanent c) processes are not classified into groups d) none of the mentioned Ans:A
24	With multiprogramming, ______________________ is used productively. a) time b) space

	c) money d) all of the mentioned Ans:A
25	The two steps of a process execution are : a) I/O & OS Burst b) CPU & I/O Burst c) Memory & I/O Burst d) OS & Memory Burst Ans:B
26	A process is selected from the ________ queue by the ________ scheduler, to be executed. a) blocked, short term b) wait, long term c) ready, short term d) ready, long term

	Ans:C
27	In the following cases non – preemptive scheduling occurs : a) When a process switches from the running state to the ready state b) When a process goes from the running state to the waiting state c) When a process switches from the waiting state to the ready state d) All of the mentioned Ans:B
28	The switching of the CPU from one process or thread to another is called : a) process switch b) task switch c) context switch d) all of the mentioned Ans:D
29	Scheduling is done so as to :

	a) increase CPU utilization b) decrease CPU utilization c) keep the CPU more idle d) None of the mentioned Ans:A
30	Scheduling is done so as to : a) increase the throughput b) decrease the throughput c) increase the duration of a specific amount of work d) None of the mentioned Ans:A
31	Turnaround time is : a) the total waiting time for a process to finish execution b) the total time spent in the ready queue c) the total time spent in the running queue d) the total time from the completion till

	the submission of a process Ans:D
32	Scheduling is done so as to : a) increase the turnaround time b) decrease the turnaround time c) keep the turnaround time same d) there is no relation between scheduling and turnaround time Ans:B
33	Round robin scheduling falls under the category of : a) Non preemptive scheduling b) Preemptive scheduling c) All of the mentioned d) None of the mentioned Ans:B
34	With round robin scheduling algorithm

	in a time shared system, a) using very large time slices converts it into First come First served scheduling algorithm b) using very small time slices converts it into First come First served scheduling algorithm c) using extremely small time slices increases performance d) using very small time slices converts it into Shortest Job First algorithm Ans:A
35	With round robin scheduling algorithm in a time shared system, a) using very large time slices converts it into First come First served scheduling algorithm b) using very small time slices converts it into First come First served

	scheduling algorithm c) using extremely small time slices increases performance d) using very small time slices converts it into Shortest Job First algorithm Ans:A
36	The FIFO algorithm : a) first executes the job that came in last in the queue b) first executes the job that came in first in the queue c) first executes the job that needs minimal processor d) first executes the job that has maximum processor needs Ans:B
37	The strategy of making processes that are logically runnable to be temporarily suspended is called : a) Non preemptive scheduling

	b) Preemptive scheduling c) Shortest job first d) First come First served Ans:B
38	Scheduling is : a) allowing a job to use the processor b) making proper use of processor c) all of the mentioned d) none of the mentioned Ans:A
39	The real difficulty with SJF in short term scheduling is : a) it is too good an algorithm b) knowing the length of the next CPU request c) it is too complex to understand d) none of the mentioned Ans:B

40	Preemptive Shortest Job First scheduling is sometimes called : a) Fast SJF scheduling b) EDF scheduling – Earliest Deadline First c) HRRN scheduling – Highest Response Ratio Next d) SRTN scheduling – Shortest Remaining Time Next Ans:D
41	One of the disadvantages of the priority scheduling algorithm is that : a) it schedules in a very complex manner b) its scheduling takes up a lot of time c) it can lead to some low priority process waiting indefinitely for the CPU d) none of the mentioned Ans:C

42	‘Aging’ is : a) keeping track of cache contents b) keeping track of what pages are currently residing in memory c) keeping track of how many times a given page is referenced d) increasing the priority of jobs to ensure termination in a finite time Ans:D
43	A solution to the problem of indefinite blockage of low – priority processes is : a) Starvation b) Wait queue c) Ready queue d) Aging Ans:D
44	Which of the following scheduling algorithms gives minimum average waiting time ? a) FCFS

	b) SJF c) Round – robin d) Priority Ans:B

Day 3 MCQ

1	Scheduling of threads are done by A. input B. output C. operating system D. memory Ans:C
2	Multiprogramming of computer system increases A. memory B. storage C. CPU utilization D. Cost Ans:C
3	Main memory of computer system is also called A. non volatile B. volatile C. reserved D. large Ans:B
4	Secondary memory of computer system is also called A. non volatile B. volatile C. reserved D. small ANSWER: A

5	To overcome the slow operating speeds of the secondary memory we make use of faster flash drives. a) True b) False Ans:A
6	The fastest data access is provided using a) Caches b) DRAM's c) SRAM's d) Registers

	Ans:D
7	The next level of memory hierarchy after the L2 cache is a) Secondary storage b) TLB c) Main memory d) Register Ans:c
8	The last on the hierarchy scale of memory devices is a) Main memory b) Secondary memory c) TLB d) Flash drives Ans:B
9	In the memory hierarchy, as the speed of operation increases the memory size also increases. a) True b) False Ans:B
10	In Priority Scheduling a priority number (integer) is associated with each process. The CPU is allocated to the process with the highest priority (smallest integer = highest priority). The problem of Starvation of low priority processes may never execute, is resolved by______. a)Terminating a procee b)aging c)mutual exclusion d)semaphore Ans:B
11	CPU performance is measured through______________. a)Throughput b)MHZ c) flaps d) none of the above Ans:A
12	Which algorithm suffers from Convoy Effect? a.FCFS b) SJF – Non preemptive c) SJF – Preemptive d.Round Robin Ans:A
13	What is the problem that arrives while using SJF algorithm? a.Deadlock b.Aging c.Starvation d.No Problem

	Ans:C
14	FCFS algorithm is implemented using a.Stack b.Tree c.Queue d.Graph Ans:C
15	‘Aging’ is : a.keeping track of cache contents b.keeping track of what pages are currently residing in memory c.keeping track of how many times a given page is referenced d.increasing the priority of jobs to ensure termination in a finite time Ans:D
16	Round robin scheduling falls under the category of : a) Non preemptive scheduling b) Preemptive scheduling c) All of the mentioned d) None of the mentioned Ans:B
17	With round robin scheduling algorithm in a time shared system, a) using very large time slices converts it into First come First served scheduling algorithm b) using very small time slices converts it into First come First served scheduling algorithm c) using extremely small time slices increases performance d) using very small time slices converts it into Shortest Job First algorithm Ans:A

MCQ’s on Interprocess communication and Deaadlock

1	Inter process communication : a) allows processes to communicate and synchronize their actions when using the same address space b) allows processes to communicate and synchronize their actions without using the same address space c) allows the processes to only synchronize their actions without communication d) none of the mentioned Ans:B

2	Message passing system allows processes to : a) communicate with one another without resorting to shared data b) communicate with one another by resorting to shared data c) share data d) name the recipient or sender of the message

	Ans:A
3	An IPC facility provides at least two operations : a) write & delete message b) delete & receive message c) send & delete message d) receive & send message Ans:D
4	The link between two processes P and Q to send and receive messages is called : a) communication link b) message-passing link c) synchronization link d) all of the mentioned Ans:A
5	Concurrent access to shared data may result in : a) data consistency b) data insecurity c) data inconsistency d) none of the mentioned Ans:C
6	A situation where several processes access and manipulate the same data concurrently and the outcome of the execution depends on the particular order in which access

	takes place is called : a) data consistency b) race condition c) aging d) starvation Ans:B
7	The segment of code in which the process may change common variables, update tables, write into files is known as : a) program b) critical section c) non – critical section d) synchronizing Ans:B
8	The following three conditions must be satisfied to solve the critical section problem : a) Mutual Exclusion b) Progress c) Bounded Waiting d) All of the mentioned Ans:D
9	Mutual exclusion implies that : a) if a process is executing in its critical section, then no other process must be executing in their critical sections b) if a process is executing in its critical section, then other

	processes must be executing in their critical sections c) if a process is executing in its critical section, then all the resources of the system must be blocked until it finishes execution d) none of the mentioned Ans:A
10	To enforce ………………….. two functions are provided enter-critical and exit-critical, where each function takes as an argument the name of the resource that is the subject of competition. A) Mutual Exclusion B) Synchronization C) Deadlock D) Starvation Ans:A
11	In ………………. only one process at a time is allowed into its critical section, among all processes that have critical sections for the same resource. A) Mutual Exclusion B) Synchronization C) Deadlock D) Starvation Ans:A
12	………………. when a process leaves a critical section and more than one process is waiting, the selection of a waiting

	process is arbitrary. A) Busy waiting is employed B) Starvation is possible C) Deadlock is possible D) All of the above Ans:B
13	Which of the following condition is required for deadlock to be possible? a) mutual exclusion b) a process may hold allocated resources while awaiting assignment of other resources c) no resource can be forcibly removed from a process holding it d) all of the mentioned Ans:D
14	A system is in the safe state if a) the system can allocate resources to each process in some order and still avoid a deadlock b) there exist a safe sequence c) all of the mentioned d) none of the mentioned Ans:A
15	The circular wait condition can be prevented by a) defining a linear ordering of resource types b) using thread

	c) using pipes d) all of the mentioned Ans:A
16	Which one of the following is the deadlock avoidance algorithm? a) banker's algorithm b) round-robin algorithm c) elevator algorithm d) karn's algorithm Ans:A
17	For effective operating system, when to check for deadlock? a) every time a resource request is made b) at fixed time intervals c) every time a resource request is made at fixed time intervals d) none of the mentioned Ans:C
18	Which one of the following is a visual (mathematical) way to determine the deadlock occurrence? a) resource allocation graph b) starvation graph

	c) inversion graph d) none of the mentioned Ans:A
19	To avoid deadlock a) there must be a fixed number of resources to allocate b) resource allocation must be done only once c) all deadlocked processes must be aborted d) inversion technique can be used Ans:A
20	With ……………………., requested resources are granted to processes whenever possible. A) deadlock detection B) deadlock deletion C) deadlock prevention D) deadlock avoidance Ans:A
21	

	Which of the following are the strategies needed for recovery once deadlock has been detected. i) Abort all deadlocked processes ii) Backup each deadlocked process to some previously defined checkpoint and restart all processes iii) Successively abort deadlocked processes until deadlock no longer exists. iv) Successively preempt resources until deadlock no longer exist. A) i, ii and iii only B) ii, iii and iv only C) i, iii and iv only D) All i, ii, iii and iv Ans:D
22	Which of the following is/are the variety of mechanisms provided by UNIX for inter-processor communication and synchronization are as follows. i) Pipes ii) Messages iii) Shared Memory iv) Main Memory v) Semaphores A) i, ii, iii and iv only B) i, ii, iii and v only C) i, iii, iv and v only D) ii, iii, iv and v only Ans:B
23	

	Which of the following are the mechanism used by the W2K executive to implement synchronization facilities. i) Process ii) Threads iii) Condition Variables iv) Mutex v) Semaphore A) i, ii, iii and iv only B) i, ii, iii and v only C) i, ii, iv and v only D) ii, iii, iv and v only Ans:C
24	In UNIX concurrency mechanisms, pipes and messages provide a means of communicating data across processes, whereas …………………. are used to trigger actions by other processes. A) Shared memory and signals B) Semaphores and shared memory C) Semaphores and signals D) Shared memory, semaphores and signals Ans:C
25	A ……………………. is a circular buffer allowing two processes to communicate on the producer consumer model. A) message B) pipe C) shared memory D) signal Ans:B
26	The schema used for deadlock ………………….. is invoking periodically to test for deadlock. A) prevention

	B) avoidance C) detection D) deletion Ans:C
27	While preventing deadlock with …………………………… needs no run-time computation since problem is solved in system design. A) request all resources B) preemption C) resource ordering D) finding safe path Ans:C
28	In ……………………… some resources, such as files, may allow multiple access for readers but only exclusive access for writers. A) Mutual Exclusion B) Hold and Wait C) Preemption D) Circular Wait Ans:A
29	For a deadlock to arise, which of the following conditions must hold simultaneously ? a) Mutual exclusion b) No preemption c) Hold and wait d) All of the mentioned

	Ans:D
30	Deadlock prevention is a set of methods : a) to ensure that at least one of the necessary conditions cannot hold b) to ensure that all of the necessary conditions do not hold c) to decide if the requested resources for a process have to be given or not d) to recover from a deadlock Ans:A

Sunbeam_Day4_OS_MCQ's

1	Physical memory is broken into fixed-sized blocks called________ a) frames b) pages c) backing store d) none of the mentioned Ans:A
2	Logical memory is broken into blocks of the same size called ___________ a) frames b) pages c) backing store d) none of the mentioned Ans:B

3	Every address generated by the CPU is divided into two parts : a) frame bit & page number b) page number & page offset c) page offset & frame bit d) frame offset & page offset Ans:B
4	The____________is used as an index into the page table. a) frame bit b) page number c) page offset d) frame offset

	Ans:B
5	The______table contains the base address of each page in physical memory. a) process b) memory c) page d) frame Ans:C
6	The size of a page is typically : a) varied b) power of 2 c) power of 4 d) none of the mentioned Ans:B
7	With paging there is no_________fragmentation. a) internal b) external c) either type of d) none of the mentioned Ans:B
8	Virtual memory allows : a) execution of a process that may not be completely in memory

	b) a program to be smaller than the physical memory c) a program to be larger than the secondary storage d) execution of a process without being in physical memory Ans:A
9	Virtual memory is normally implemented by_________ a) demand paging b) buses c) virtualization d) all of the mentioned Ans:A
10	The valid – invalid bit, in this case, when valid indicates : a) the page is not legal b) the page is illegal c) the page is in memory d) the page is not in memory Ans:C
11	A page fault occurs when : a) a page gives inconsistent data b) a page cannot be accessed due to its absence from memory c) a page is invisible d) all of the mentioned Ans:B
12	In segmentation, each address is specified by : a) a segment number & offset

	b) an offset & value c) a value & segment number d) a key & value Ans:A
13	In paging the user provides only__________which is partitioned by the hardware into___________and_______ a) one address, page number, offset b) one offset, page number, address c) page number, offset, address d) none of the mentioned Ans:A
14	Each entry in a segment table has a : a) segment base b) segment peak c) segment value d) none of the mentioned Ans:A
15	The segment base contains the : a) starting logical address of the process b) starting physical address of the segment in memory c) segment length d) none of the mentioned

	Ans:B
16	The segment limit contains the : a) starting logical address of the process b) starting physical address of the segment in memory c) segment length d) none of the mentioned Ans:C
17	______is the concept in which a process is copied into main memory from the secondary memory according to the requirement. a) Paging b) Demand paging c) Segmentation d) Swapping Ans:B
18	A process is thrashing if a) it is spending more time paging than executing b) it is spending less time paging than executing c) page fault occurs d) swapping can not take place

	Ans:A
19	The three major methods of allocating disk space that are in wide use are : a) contiguous b) linked c) indexed d) all of the mentioned Ans:D
20	In contiguous allocation : a) each file must occupy a set of contiguous blocks on the disk b) each file is a linked list of disk blocks c) all the pointers to scattered blocks are placed together in one location d) none of the mentioned Ans:A
21	In linked allocation : a) each file must occupy a set of contiguous blocks on the disk b) each file is a linked list of disk blocks c) all the pointers to scattered blocks are placed together in one location d) none of the mentioned

	Ans:B
22	In indexed allocation : a) each file must occupy a set of contiguous blocks on the disk b) each file is a linked list of disk blocks c) all the pointers to scattered blocks are placed together in one location d) none of the mentioned Ans:C
23	Contiguous allocation of a file is defined by : a) disk address of the first block & length b) length & size of the block c) size of the block d) total size of the file Ans:A
24	________and__________are the most common strategies used to select a free hole from the set of available holes. a) First fit, Best fit b) Worst fit, First fit c) Best fit, Worst fit d) None of the mentioned

	Ans:A
25	For each file there exists a_____________that contains information about the file, including ownership, permissions and location of the file contents. a) metadata b) file control block c) process control block d) all of the mentioned Ans:B
26	Metadata includes : a) all of the file system structure b) contents of files c) both file system structure and contents of files d) none of the mentioned Ans:C
27	In the linked allocation, the directory contains a pointer to the : I. first block II. last block a) I only b) II only c) Both I and II

	d) Neither I nor II Ans:C
28	There is no____________with linked allocation. a) internal fragmentation b) external fragmentation c) starvation d) all of the mentioned Ans:B
29	The major disadvantage with linked allocation is that : a) internal fragmentation b) external fragmentation c) there is no sequential access d) there is only sequential access Ans:D
30	Contiguous allocation has two problems____________and____________ that linked allocation solves. a) external – fragmentation & size – declaration b) internal – fragmentation & external – fragmentation c) size – declaration & internal – fragmentation d) memory – allocation & size – declaration

	Ans:A
31	Each ________has its own index block. a) partition b) address c) file d) all of the mentioned Ans:C
32	Indexed allocation____________direct access. a) supports b) does not support c) is not related to d) none of the mentioned Ans:A
33	A memory page containing a heavily used variable that was initialized very early and is in constant use is removed, then the page replacement algorithm used is : a) LRU b) LFU c) FIFO d) None of the mentioned

	Ans:C
34	The aim of creating page replacement algorithms is to : a) replace pages faster b) increase the page fault rate c) decrease the page fault rate d) to allocate multiple pages to processes Ans:C
35	A FIFO replacement algorithm associates with each page the ________ a) time it was brought into memory b) size of the page in memory c) page after and before it d) all of the mentioned Ans:A
36	Optimal page – replacement algorithm is : a) Replace the page that has not been used for a long time b) Replace the page that has been used for a long time c) Replace the page that will not be used for a long time d) None of the mentioned Ans:C
37	Optimal page – replacement algorithm is difficult to implement,

	because : a) it requires a lot of information b) it requires future knowledge of the reference string c) it is too complex d) it is extremely expensive Ans:B
38	In the________algorithm, the disk arm starts at one end of the disk and moves toward the other end, servicing requests till the other end of the disk. At the other end, the direction is reversed and servicing continues. a) LOOK b) SCAN c) C-SCAN d) C-LOOK Ans:B
39	In the_________algorithm, the disk head moves from one end to the other , servicing requests along the way. When the head reaches the other end, it immediately returns to the beginning of the disk without servicing any requests on the return trip. a) LOOK b) SCAN c) C-SCAN d) C-LOOK

	Ans:C
40	In the________algorithm, the disk arm goes as far as the final request in each direction, then reverses direction immediately without going to the end of the disk. a) LOOK b) SCAN c) C-SCAN d) C-LOOK Ans:A